EHR Clinical Intelligence & Interoperability Platform

Master Technical Specification & Engineering Artifacts: Weeks 1 & 2

Executive Technical Summary: *This consolidated master document specifies the complete architectural, algorithmic, persistence, and workflow implementations delivered across Week 1 (Multi-Source Ingestion & Clinical NLP) and Week 2 (Patient Identity Resolution EMPI, Text Fusion & Doctor Review UI). All data models, scoring functions, safety invariants, and test results are verified against running services and canonical database stores.*

EHR Clinical Intelligence Platform

Week 1 Technical Artifacts: Multi-Source Heterogeneous Ingestion, Canonical Data Models & Clinical NLP Baseline

Author: AI Clinical Engineering Team	**Status: Verified & In Production**	**Generated: 2026-09-18**
Target Services: Ingestion (8000), Terminology (8001)	**Infrastructure: PostgreSQL (5432, 5433), MinIO S3 (9000)**	**Test Suite: 6/6 Integration Tests Passing**

1. System Topology & Core Infrastructure

Week 1 deployed the foundational persistence layer separating operational EHR transactions from high-dimensional terminology vector lookups. All components run as containerized microservices managed through Docker Compose meshes.

Component	Technology	Port	Schema / Bucket / Role
Primary Operational DB	PostgreSQL 16	`5432`	Database: ehr_primary (patients, encounters, documents, fragments)
Terminology Vector DB	PostgreSQL 16 + pgvector	`5433`	Database: terminology_db (384-dim dense cosine embeddings)
Object Storage Engine	MinIO S3 API	`9000 / 9001`	Bucket: ehr-raw-inputs (SHA-256 verified raw payloads)
Ingestion Microservice	FastAPI / Python 3.11	`8000`	REST API: Multi-adapter ingestion, seeding, queries
Terminology Microservice	FastAPI / Python 3.11	`8001`	Hybrid lexical & vector semantic search over ontologies

2. Canonical Relational Models (services/ingestion/models.py)

The relational schema decouples heterogeneous input streams into normalized entities with full foreign key constraints and audit indexes.

```
-- Canonical Source Document Catalog
CREATE TABLE source_documents (
    id VARCHAR PRIMARY KEY,               -- e.g. STT-2026-081234
    patient_id VARCHAR REFERENCES patients(id), -- NULL if quarantined
    encounter_id VARCHAR REFERENCES encounters(id),
    source_type VARCHAR NOT NULL,          -- SPEECH_TO_TEXT, OCR, HISTORICAL_DOCUMENT, HL7_API_JSON
    document_class VARCHAR,                -- consultation_transcript, prescription
    minio_path VARCHAR NOT NULL,           -- S3 URI in ehr-raw-inputs
    sha256_checksum VARCHAR(64) NOT NULL,  -- 64-char hexadecimal digest
    raw_text TEXT NOT NULL,
    identity_hint_fields JSONB,            -- Demographic hints for EMPI
    status VARCHAR DEFAULT 'INGESTED'      -- Lifecycle: INGESTED, PENDING_IDENTITY_RESOLUTION, LINKED
);

-- Atomic Clinical Fragments
CREATE TABLE source_fragments (
    id VARCHAR PRIMARY KEY,
    document_id VARCHAR REFERENCES source_documents(id) ON DELETE CASCADE,
    fragment_index INTEGER NOT NULL,
    text_content TEXT NOT NULL,
    confidence FLOAT DEFAULT 1.0,
    speaker_turn VARCHAR,                  -- DOCTOR | PATIENT
    bounding_box JSONB                     -- [ymin, xmin, ymax, xmax]
);
```

3. Heterogeneous Multi-Source Ingestion Adapters

The ingestion engine supports 4 distinct clinical stream types, each handled by an isolated adapter in services/ingestion/adapters/:

Adapter	Source Format	Extraction Logic & Attributes	Quarantine Policy
STTAdapter	Diarized Audio JSON	Extracts speaker turns (DOCTOR/PATIENT), start/end millisecond timestamps, acoustic confidences.	Immediate Link (status = INGESTED)
OCRAdapter	Scanned Prescriptions	Extracts text blocks, OCR vision confidence scores, and bounding box coordinates [ymin, xmin, ymax, xmax].	Current Encounter (status = INGESTED)
HistoricalAdapter	Legacy Hospital PDFs / OCR	Extracts demographic hints (Name, DOB, Gender, MRN, Facility) into identity_hint_fields.	Strict Quarantine (status = PENDING_IDENTITY_RESOLUTION, patient_id = None)
HL7APIAdapter	FHIR / HL7 v2 JSON	Unpacks Condition, Observation, MedicationRequest, and DiagnosticReport resource collections.	Immediate Link (status = INGESTED)

> ***Architectural Safety Invariant:*** *Per EHR Architecture Sec 4.3, external historical records are strictly quarantined on arrival. They are assigned patient_id = None and status = 'PENDING_IDENTITY_RESOLUTION' to guarantee zero leakage into the active patient chart prior to verified identity resolution.*

4. Clinical AI & NLP Baseline Extractor

Implemented in services/clinical-processing/nlp/extractor.py, the engine performs multi-class clinical entity extraction, assertion analysis (NegEx scope resolution), and temporality classification.

- **Multi-Class NER:** Extracts MEDICATION (agents, formulations), DOSAGE (e.g. 500mg), FREQUENCY (e.g. twice daily with meals), DIAGNOSIS (e.g. Type 2 Diabetes Mellitus, Acute Pancreatitis), and LAB_OBSERVATION (e.g. HbA1c 8.2%, Lipase

210 U/L).

- **Negation Resolution (NegEx):** Evaluates a forward character window of 40 characters from trigger tokens ('denies', 'no history of', 'negative for', 'ruled out'). Extracted entities within the scope are assigned assertion_status = NEGATED.
- **Temporality Categorization:** Discriminates between HISTORICAL diagnoses ('history of', 'resolved', 'prior') and ACTIVE conditions ('current', 'active', 'taking').

5. Week 1 Verification Suite (tests/test_week1_pipeline.py)

Test Identifier	Scope & Assertions	Observed Output	Result
test_01_minio_connectivity	MinIO S3 connection, bucket creation, SHA-256 hash length == 64, payload round-trip.	`URI: ehr-raw-inputs/test/test-pipe-01.json, SHA256 verified`	**PASS (ok)**
test_02_ingest_speech_to_text	Ingest audio transcript, verify doc ID, status == INGESTED, 6 diarized fragments.	`doc.id == STT-2026-081234, len(fragments) == 6`	**PASS (ok)**
test_03_ingest_ocr_prescription	Ingest scanned prescription, verify document_class, bounding_box persistence.	`doc.id == OCR-2026-081234, bounding_box verified`	**PASS (ok)**
test_04_ingest_historical_doc	Verify quarantine invariant: status == PENDING_IDENTITY_RESOLUTION, patient_id == None.	`Quarantined doc with extracted Rohan Sharma hints`	**PASS (ok)**
test_05_ingest_hl7_api_json	Unpack FHIR/HL7 collection bundle into discrete clinical observation fragments.	`doc.id == HL7API-2026-081234, 6 fragments created`	**PASS (ok)**
test_06_nlp_extraction	Execute NER extractor over ingested narrative, verify MEDICATION and DIAGNOSIS extraction.	`entity_count > 0, MEDICATION and DIAGNOSIS extracted`	**PASS (ok)**

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Week 2 Technical Artifacts: Hybrid Patient Identity Resolution, Text Fusion & Dual-Theme Review UI

Author: AI Clinical Engineering Team	**Status: Verified & In Production**	**Generated: 2026-09-18**
Target Services: EMPI (8002), Fusion (8003), UI (5173)	**Algorithms: Jaro-Winkler, Phenotype Jaccard, Safety Vetoes**	**Test Suite: 6/6 Integration Tests Passing**

1. Hybrid Multi-Factor Patient Identity Resolution (EMPI)

Operating on Port 8002 (services/identity-resolution/), the EMPI engine associates unindexed external documents with candidate patients. It combines multi-factor demographic string distances, clinical phenotype ontology overlap, and hard safety vetoes.

```
COMPOSITE MATCH SCORING FORMULATION:
  S_total = (0.85 * S_demo) + (0.15 * S_pheno)

Where Demographic Score (S_demo) is a weighted sum:
  S_demo = 0.30*S_name + 0.25*S_dob + 0.10*S_gender + 0.15*S_nat_id + 0.10*S_phone + 0.10*S_addr
  • S_name: Jaro-Winkler distance with token sort ratio fallback
  • S_dob: Exact ISO match (1.0), transposition tolerance (0.80), year-month (0.70)
  • S_nat_id: Masked trailing suffix match (e.g. XXXX-XXXX-4432 -> 4432)
  • S_phone: Trailing 10-digit unmasked alignment

Where Phenotype Overlap (S_pheno) evaluates SNOMED CT and RxNorm concept overlap:
  S_pheno = Jaccard(C_patient, C_doc) = |C_patient n C_doc| / |C_patient u C_doc|
```

Deterministic Clinical Safety Vetoes:
*1. **Birth Year Discrepancy:** If |Year(DOB_doc) - Year(DOB_patient)| > 2 years -> FATAL VETO.*
*2. **Biological Gender Conflict:** If Gender_doc != Gender_patient (both specified) -> FATAL VETO.*
***Consequence:** Regardless of name similarity, S_total is capped at <= 0.35, tier = MANUAL_REVIEW, and the document is permanently quarantined.*

Confidence Decision Tiers & Forensic Audit Trail

Confidence Tier	Score Range	Decision Action	Clinical Routing & Pipeline Behavior
Tier 1 (High)	`Score >= 0.85`	**AUTO_LINK**	Automatically assigns patient_id in database, updates status = 'LINKED'. Enters patient timeline immediately.
Tier 2 (Medium)	`0.60 <= Score < 0.85`	**DOCTOR_VERIFICATION**	Holds document in quarantine (status = 'PENDING_IDENTITY_RESOLUTION'). Dispatches to physician review queue.
Tier 3 (Low / Veto)	`Score < 0.60 or Veto`	**MANUAL_REVIEW**	Quarantined permanently (status = 'QUARANTINED'). Completely excluded from clinical timeline.

2. Clinical Text Normalization & Source Fusion (Port 8003)

Implemented in services/clinical-processing/fusion/engine.py, the fusion engine normalizes heterogeneous fragments into a unified patient trajectory.

- **Strict Quarantine Invariant:** Strictly queries documents WHERE status IN ('INGESTED', 'LINKED', 'PROCESSED'). Any document in PENDING_IDENTITY_RESOLUTION, QUARANTINED, or REJECTED status is physically excluded, guaranteeing ZERO TIMELINE LEAKAGE.
- **Chronological Patient Trajectory:** Interleaves multi-modal records (audio consultations, prescriptions, lab feeds, resolved external discharge summaries) sorted chronologically by UTC timestamp.
- **Cryptographic Provenance Verification:** Every timeline item and narrative assertion embeds its MinIO S3 URI and 64-character SHA-256 checksum.
- **Synthesized Working Narrative:** Organizes patient records into 4 structured sections: (1) Active Consultation Dialogue, (2) Diagnostic/Lab Observations, (3) Reconciled Historical Precedents, (4) Cryptographic Provenance Ledger.

3. Doctor Review UI & Dual-Theme Monochrome Architecture

Built with React 19 + TypeScript + Vite on Port 5173, the web application delivers a high-contrast, minimalist clinical interface.

Clinical View	Route	Primary Capabilities & Physician Workflows
Patient Dashboard	`/`	Real-time stream feed across all 4 clinical sources, encounter banner, fragment inspector, raw JSON viewer.
Verification Queue	`/review`	Side-by-side comparison of incoming document hints vs registered candidate patient, multi-factor score breakdown meters, one-click [Confirm Link] and [Reject & Quarantine] actions.
Unified Timeline	`/timeline`	Patient selector dropdown, vertical chronological timeline with 1px hairline guide and node dots, copyable SHA-256 hash, and synthesized clinical working narrative.

Dual-Theme Inverted Architecture (Bright Mode vs Dark Mode)

Per strict clinical design specifications, the UI adheres to a **zero-rainbow color palette** (no blues, greens, ambers, or purples), employing semantic CSS variables for instant inversion:

Semantic Token	Dark Mode ([data-theme='dark'])	Bright Mode ([data-theme='light'])	Clinical Intent
`--bg-main`	Deep Void Black (#000000)	Stark Pure White (#ffffff)	Full canvas base
`--bg-surface`	Onyx (#0a0a0a)	Off-White Card (#f7f7f8)	Card container elevation
`--text-primary`	Pure White (#ffffff)	Solid Black (#000000)	Primary clinical typography
`--code-bg`	Pure Black (#000000)	Soft Code Gray (#f2f2f5)	Monospace narrative container
`--btn-primary-bg`	Solid White (#ffffff)	Solid Black (#000000)	High-contrast action trigger
`--timeline-dot`	White with black border	Black with white border	Chronological event marker

4. Week 2 Verification Suite (tests/test_week2_pipeline.py)

Test Identifier	Scope & Assertions	Observed Output	Result
test_01_tier1_autolink	High confidence (>= 0.85) auto-associates patient and updates status to LINKED.	`tier == AUTO_LINK, score >= 0.85, doc.status == LINKED`	**PASS (ok)**
test_02_tier2_doctor_queue	Medium confidence (0.60 - 0.84) holds in quarantine and routes to review queue.	`tier == DOCTOR_VERIFICATION, present in review queue`	**PASS (ok)**
test_03_doctor_confirmation	Physician confirms linkage: document transitions to LINKED, exits queue, updates DB.	`confirm success, doc linked to PT-HOSP-MRN-55210`	**PASS (ok)**
test_04_tier3_hard_veto	Birth year discrepancy > 2 years triggers hard veto, caps score <= 0.40, quarantines.	`tier == MANUAL_REVIEW, FATAL CONFLICT detected, status == QUARANTINED`	**PASS (ok)**
test_05_phenotype_boost	Matching SNOMED CT and RxNorm concepts verified to boost overall linkage confidence.	`matching score > irrelevant score, overlap verified`	**PASS (ok)**
test_06_quarantine_isolation	Quarantined docs strictly excluded from timeline; cleared docs sorted chronologically.	`Quarantined doc absent from timeline; UTC order verified`	**PASS (ok)**